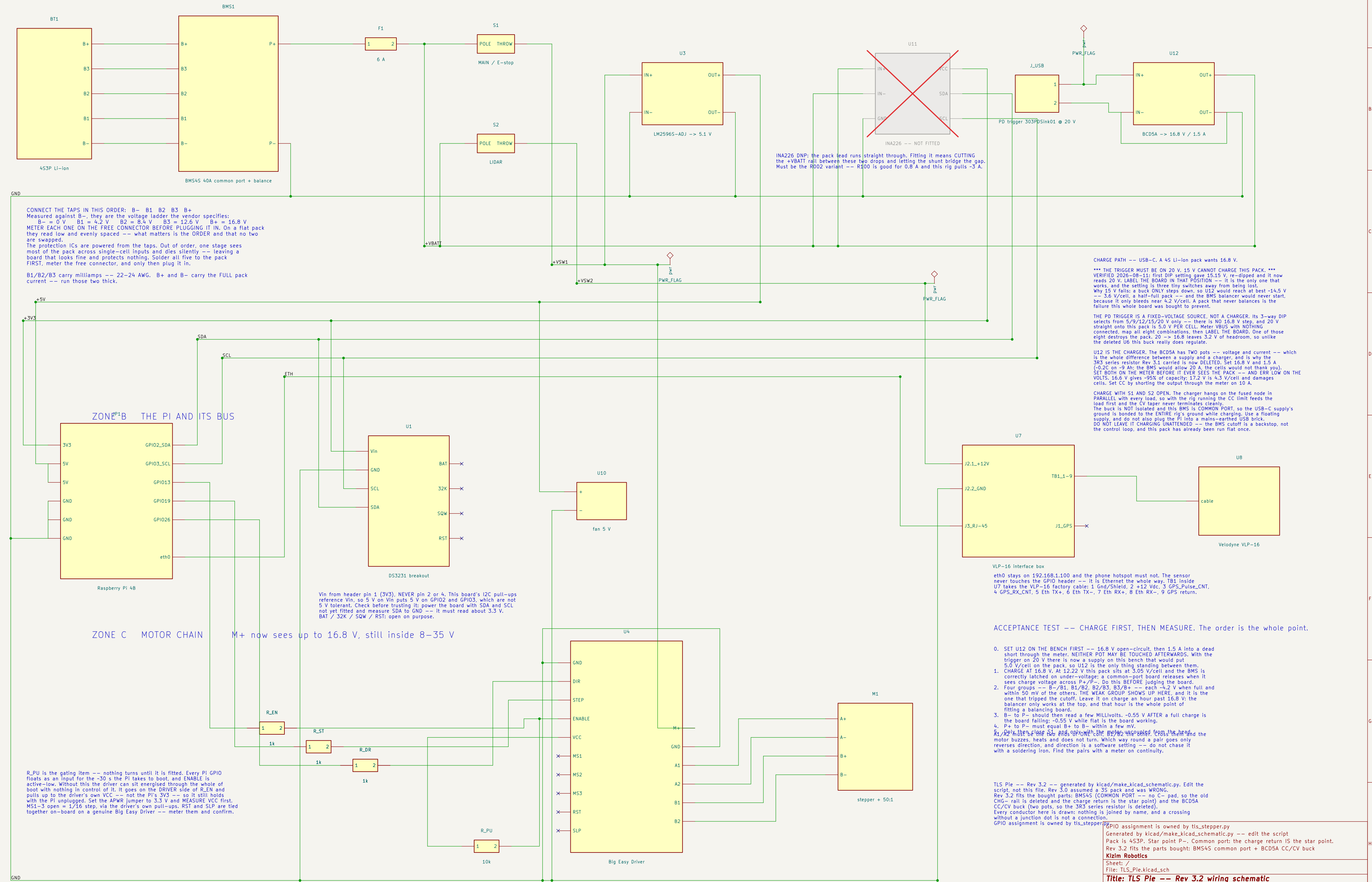


ZONE A POWER CHAIN 4S3P pack, common-port balancing BMS, USB-C charge

S1 and S2 both hang off the fused node, in PARALLEL. S1 does NOT switch the lidar -- opening S1 kills the Pi and the motor and leaves the VLP-16 spinning on S2.

*** NEW ON 4S: S2 HANDS THE VLP-16 UP TO 16.8 V ***
On the old 3S assumption that leg never passed 12.6 V. CHECK THE SENSOR'S OWN INPUT RANGE before S2 is ever closed. If it will not take 16.8 V that leg needs a regulator -- the one place going to 4S makes things worse.



ACCEPTANCE TEST -- CHARGE FIRST, THEN MEASURE. The order is the whole point.

0. SET U12 ON THE BENCH FIRST! — 16.8 V open-circuit, then 1.5 A into a dead short through the meter. NEITHER POT MAY BE TOUCHED AFTERWARDS. With the rig set up, the 16.8 V is the voltage across the 100 ohm resistor. The 1.5 A is 5.0 V/cell on the pack, so U12 is the only thing standing between them. Charge at 16.8 V. On U12 this 16.8 V pack sits at 3.05 V/cell and the BMS is correctly charged on the 1.5 A. The 1.5 A is the current through the battery when it sees charge voltage across $P+P'$. DO NOT BEFORE judging the board.
 2. Four groups — $B- / B1, B2, B2/B3, B3/B4 -$ each ~ 4.2 V when full and within 50 mV of the others. THE WEAK GROUP SHOWS UP HERE, and is the one that is charged the most. The 1.5 A is the current through the battery when the balancer only works at the top, and that hour is the whole point of fitting a balancing board.
 3. $B- = P+$ should then be a few MILLIVOLTS. -0.55 V AFTER a full charge is the board falling: -0.55 V while flat is the board working.
 4. $P+ = P-$ must equal $B+ = B-$ within a few mV.
- At the end of the test: The battery is fully charged, the BMS is UNCHARGED, and the motor buzzes, heats and does not turn. Which way round a pair goes will be obvious, and the direction of the softest buzz will tell you which to chase it with a soldering iron. Find the pairs with a meter on continuity.

TIS Pie -- Rev 3.2 -- generated by kicad/make_kicad_schematic.py. Edit the script, not this file. Rev 3.0 assumed a 35 pack and was WRONG.
 Rev 3.2 fits the bought parts: BM545 (COMMON PORT --- no C- pad, so the o CHG- rail is deleted and the charge return is the star point) and the BCD5A CC/CV buck (two pots, so the 3R3 series resistor is deleted).
 Every conductor here is drawn: nothing is joined by name, and a crossing without a junction dot is not a connection.
 GPIO assignment is owned by tIs-stepperpilot. GPIO assignment is owned by tIs-

GPIO assignment is owned by `tls_stepper.py`
Generated by `kiCad/make_kiCad_schematic.py` -- edit the script
Pack is 4S3P. Star point P-. Common port: the charge return IS the star point.

Rev 3.2 fits the parts bought: BMS4S common port + BCD5A CC/CV buck
Kizim Robotics

Sheet: /
File: TLS_Pie.kicad_sch

Title: TLS Pie -- Rev 3.2 wiring schematic

Size: A2	Date: 2026-08-11	Rev: 3.
----------	------------------	---------